

Sky Pemberton

MA346-1

Due 12/10/24

What are the most effective strategies for improving employee retention?

Introduction

Employee retention is an integral part to an organizations success, increasing productivity, reducing costs, and overall providing a positive and collaborative workplace. This report discusses why employees leave or are terminated from a company, and how to improve retention, with factors such performance, employee satisfaction, and managerial influence. Contrary to common assumption, salary is not a significant factor in employee attrition, although it is also analyzed.

During this report, the methods and data section will discuss the dataset, data cleaning, and other choices made to help provide analysis. The next section, results, will present the findings from analysis, followed by the discussion section which will contextualize and interpret the findings. Lastly will be the Conclusion, which will sum up the important takeaways.

Methods and Data

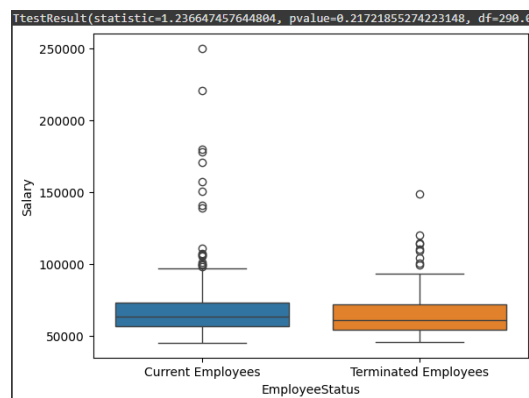
The dataset, from Kaggle.com, contains 311 employees from a company, each summarized by 36 values providing information on their performance, demographic, retention, and more. The average salary is around \$69k, with a range between \$45k-\$250k. 33% of the dataset contains

ex-employees, and it provides a quick explanation of why they were fired or left. There is also data on managers, and there are 23 managers that manage the 311 employees. Average employee satisfaction is rated on a scale 1-5, is 3.89. This average shows that most employees are satisfied with the company. Lastly, employee engagement and performance is also provided, with engagement scores averaging 4.11 on a 5-point scale. This shows that while most employees are engaged, there are areas where improvements can be made.

The two main ways an employee leaves a company is because they are terminated, or they leave voluntarily. In the data, there are reasons for why employees leave that have nothing to do with the company, like school, military, or relocation. I have decided to remove that data to better focus on what the company can do to improve attrition. This decreases the dataset size to 292 employees with 22 managers, so our sample size isn't lessened too much.

Results

Figure 1



A large factor in whether an employee decides to stay is compensation, which oftentimes makes individuals leave even if they are satisfied with everything else at the company. This can

result in a trend where lower paid employees leave at a higher rate than higher paid employees. At this company, however, salary did not seem to play much of a role, denoted in figure 1. There was no statistically significant difference between current and terminated employee salaries, although the highest paid employees at the company are all still there. When analyzing salaries by department, the Admin Offices department seems to have the greatest difference in salary (Figure 2), but then showed no significant increase in termination rate within the department (Figure 3).

Figure 2

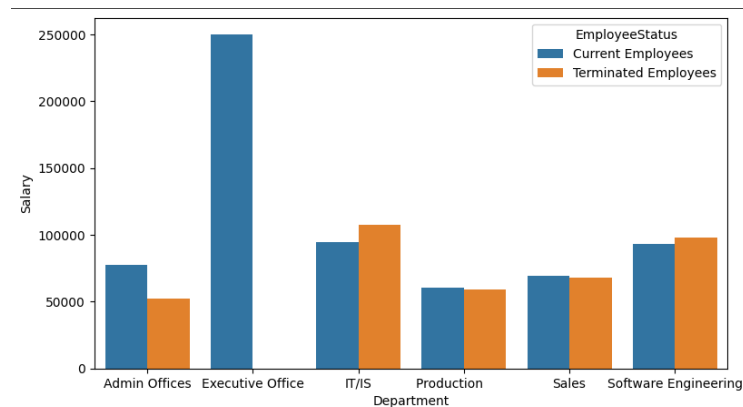
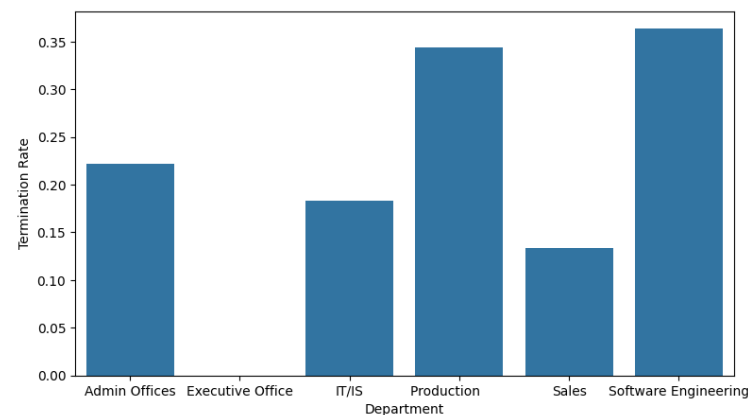
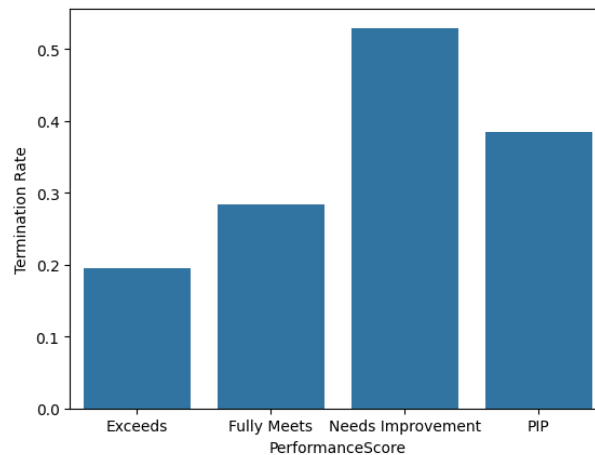


Figure 3



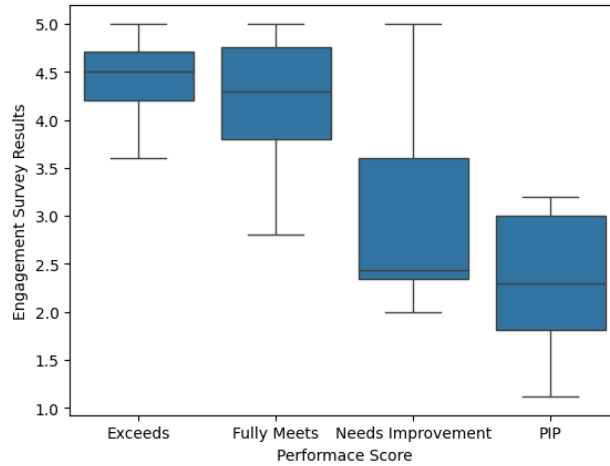
While salary does not appear to be a primary driver of termination, performance scores can show if employees are meeting expectations. The relationship between performance and termination rate (*Figure 4*) shows that half of employees who ‘need improvement’ end up being terminated. Additionally, almost 40% of employees on performance improvement plans (PIP) are also terminated from the company. This result supports the conclusion that poor performing employees count for a large proportion of attrition, but does not explain why these employees are failing, and what the company can do about it.

Figure 4



Further analyzing performance score, *Figure 5* shows how these employees respond to the engagement survey, which is below average. Under performing employees do not feel engaged in the workplace, which explains why many leave the company.

Figure 5



One reason for poor engagement could be explained by management, with some managers having lower average survey results than others. Amy Dunn, and Kissy Sullivan, two production department managers, have low average survey results (*Figure 6*). They also happen to have the highest subordinate termination rate (*Figure 7*), which may show that poor management in certain departments is a potential reason for attrition. The production department had one of the highest termination rates (*Figure 3*), and this is likely to explain why. The company could see less employee turnover by hiring new production managers that promote better engagement with their subordinates.

Figure 6

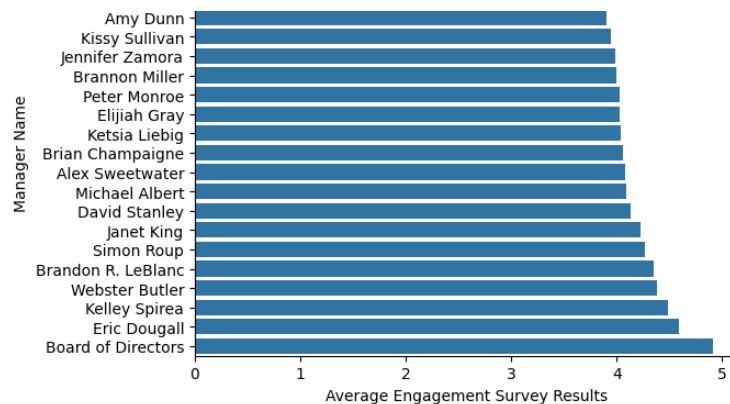
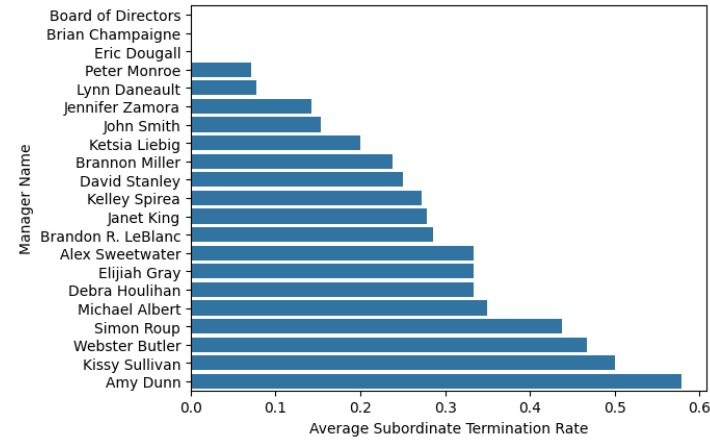


Figure 7



Discussion

The results of this analysis provide insights explaining employee terminations and highlight factors that influence turnover. Many organizations will often try to increase salaries to keep employees from leaving, but this analysis shows that it is not a significant factor which will save them money. The company should instead shift its attention to addressing employee engagement and managerial factors.

Underperforming employees often are terminated within the company, but they need to make sure that they understand why these employees are struggling. If there are a lot of employees reporting poor engagement and leaving who have the same manager, then they should investigate management to try and fix the problem. Improving management training, or just replacing ineffective managers could increase employee retention.

In *Figure 3*, the software engineering department had the highest termination rate, but upon further investigation there was no obvious reason why rates were so high. Performance scores, engagement surveys, and employee satisfaction were all average or above average in the data, so this most likely can be explained by outside factors. Some of the termination reasons for this department included poor attendance, career changes, and switching positions, which could be explained by the environments that these employees work in. Software engineering is a high-pressure and mentally demanding field that often comes with stress and tight deadlines, which could explain the increased termination rate. To help combat this, the company may want to implement a better work life balance or stress management training in this department which can help decrease termination rates.

Conclusion

This data analysis provided insights into the factors influencing attrition within a company and identified areas where improvements could be made. Performance scores revealed a clear relationship with attrition, but explaining why employees were underperforming is more important. Promoting engagement in the workplace can help improve performance, and in addition they should consider ineffective management. Replacing management could improve employee engagement, which will then improve performance and decrease turnover. Lastly, encouraging work-life balance and reducing stress in the software engineering department could help address high attrition rates in this area.

By resolving these key areas, engagement, performance, and management quality, the company can reduce attrition and create a more supportive and productive work environment.